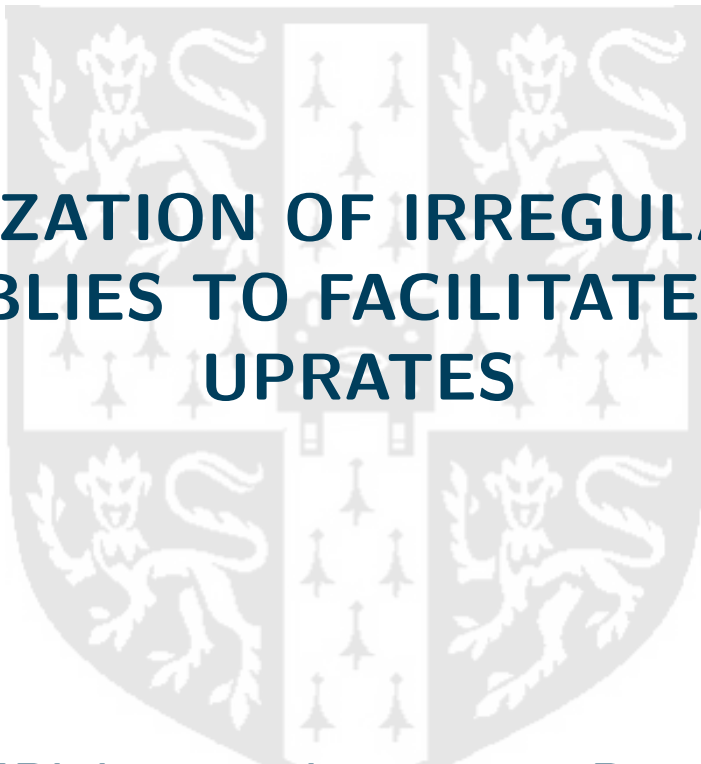




OPTIMIZATION OF IRREGULAR PWR ASSEMBLIES TO FACILITATE POWER UPDATES

MPhil research project - Report

Supervisor : Professor Geoff Parks

Léa Girard



UNIVERSITY OF
CAMBRIDGE

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1

INTRODUCTION

1.1 CONTEXT AND MOTIVATION

As the world's interest in nuclear power grows and new build proves to be difficult and costly, power uprates can be an important option to leverage. A power uprate is the increase of a reactor's power after it has first been commissioned. This allows the utility to provide additional nuclear power to the grid with less associated risks than a new build.

Numerous power uprates have indeed been implemented in the US : the U.S. Nuclear Regulatory Commission (US NRC) has approved more than 144 power uprates since 1977 [1], totalling to 24.080 MW thermal in 2021. Uprates have also been conducted outside of the US, for example EDF in France uprated some of its 900 MW reactors through replacement of the low pressure turbine [2].

There are different types of power uprates, as stated by the US NRC [3] [4] :

- **Measurement uncertainty recapture power uprates.** These are usually less than 2% increases achieved through measurement uncertainty recapture. This is done by improving instrumentation in the reactor, especially core power measurement.
- **Stretch power uprate.** These can achieve up to 7% uprates through stretching the plant's design basis with only minimal changes to the plant. It amounts to running the plant more aggressively by reducing operating margins.
- **Extended power uprates.** These include any uprate above 7% and have been granted in the US up to 20%. This type of uprate involves large work done on the plant, usually upgrading the turbo generator and other systems playing an important part in efficiency, such as pumps. It is however a complicated and expensive process.

New power uprates Researchers are interested in other types of power uprates, to achieve greater power increases for existing reactors. In this work, we will focus on traditional pressurized water reactors (PWRs).

The main option that has been studied for more important power uprates than the three options listed above is to change the fuel. Indeed, this allows for bigger changes in power while maintaining all safety parameters. However, no such uprate has been put in place yet.

Many different avenues have been explored for such power uprates, such as annular fuel with internal cooling [5] [6], accident tolerant fuel [7], fuel doped to increase its density [8], and increasing the number of fuel pins.

This latter solution is the one we will study in this project. Indeed, reducing pin diameter allows for reduced temperatures of fuel and cladding, which allows for increased power while maintaining safety margins [7].

Some research [9] has already been done on increasing the number of fuel pins; however, the solutions proposed reworked the position of guide tubes, which made power uprates of reactors currently in operation prohibitively expensive.

One paper by Ben Lindley [10] has studied backwards-compatible assemblies for PWRs allowing for power uprates through a larger number of smaller fuel pins, compared to the traditional 17x17 design.

1.2 AIM OF THIS WORK

Ben Lindley has demonstrated in [10] that a backwards compatible assembly with smaller and more numerous fuel pins can allow for a power uprate by increasing margins to safety parameters like fuel temperature. However, his choice of design for a given number of fuel pins was made through his own reasoning and intuition.

The goal of this project is to apply formal optimization methods to this problem and find optimal positions for the new more numerous fuel pins in the assembly.

2

PREVIOUS WORK

The design proposed by Lindley in [10] aims to be backwards compatible. Thus, it respects the assembly size and guide tube size and position of current PWR assemblies to ensure compatibility with current pressure vessel perforations. This is achieved by allowing the position of fuel pins to be irregular : the pins do not follow a square lattice like in the 17x17 design. Instead, each pin is positioned independently rather than following an overall structure.

The fuel and cladding are traditional, in order to use existing safety tests, experimental measurements of important values such as heat transfer parameters, and operational experience from the current fleet.

2.1 REQUIREMENTS FOR THE NEW ASSEMBLIES

When designing the new assemblies to facilitate power uprates, Lindley chose a set of requirements that could make the new design feasible, safe and backwards compatible with current PWRs. In this project we also follow these requirements.

- The assembly outer dimensions and guide tube positions must not be changed, in order to ensure backwards compatibility of pressure vessel perforations.

- The hydrogen to heavy metal (H/HM) ratio is conserved in order to maintain the agreed-upon optimum of under moderation needed for negative feedback while also keeping a good neutron economy.
- Reactivity and the evolution of reactivity with burnup should not be much worse than in current assemblies to ensure operability of the reactor and economic viability. The beginning of life reactivity should follow directly from the H/HM ratio and thus be similar to current assemblies. Still, a burnup simulation has been carried out by Lindley to check for cycle length. However, any deficiency regarding this requirement could be compensated with higher enrichment, so it is not a large issue.
- There is a minimum space to respect between pins to allow for structural supports along the assembly. The limit chosen by Lindley is 2 mm. This limit was applied only between fuel pins and not with respect to guide tubes.
- The minimum diameter of a fuel pin is determined to prevent vibration. Lindley chose it at 7 mm.
- The minimum departure from nucleate boiling ratio (MDNBR) must not be much lower than in current assemblies to ensure safety.
- The control rod worth must not be much lower than in current assemblies to ensure safety by maintaining the shutdown margin. It is possible to get around this requirement since it could be possible to modify control rods in a PWR.
- The heating of the coolant within the assembly should be quite even. There is a margin allowed since the coolant mixes above the core.
- The pressure drop of coolant going through the assembly should not differ too much from current assemblies. Indeed, when the new assembly design is phased in the cores will be mixed between traditional and new assemblies, and a pressure difference could lead to some assemblies lacking sufficient flow. For a stable uprated core that should not be an issue since the pumps would have to be updated for the new power output anyway.

Lindley poses two last requirements that we will choose to relax.

First, the heat flux of the different pins should be reasonably uniform to ensure even heating of the coolant. This is necessary to prevent the MDNBR from being eroded. This leads to forcing the pins to all have the same size, to keep pin powers similar. While we will indeed start the project studying pins of a single radius, this requirement could be relaxed in the future since we are modelling MDNBR and power peaking directly. However, practical considerations also need to be taken into account, so having more than two sizes of pins seems unreasonable.

Secondly, Lindley poses the requirement that the within assembly power peaking should be limited in order to prevent high fuel temperatures and low MDNBR. However, this is not a binding requirement. In the following reasoning we will consider power peaking as a loss of the power uprate coming from the larger number of pins rather than a limit.

2.2 CONCLUSION REACHED

Design Lindley concludes his paper by presenting an assembly that has demonstrated a possible power uprate. It has 324 pins instead of the usual 264. The design is shown in Figure 1.

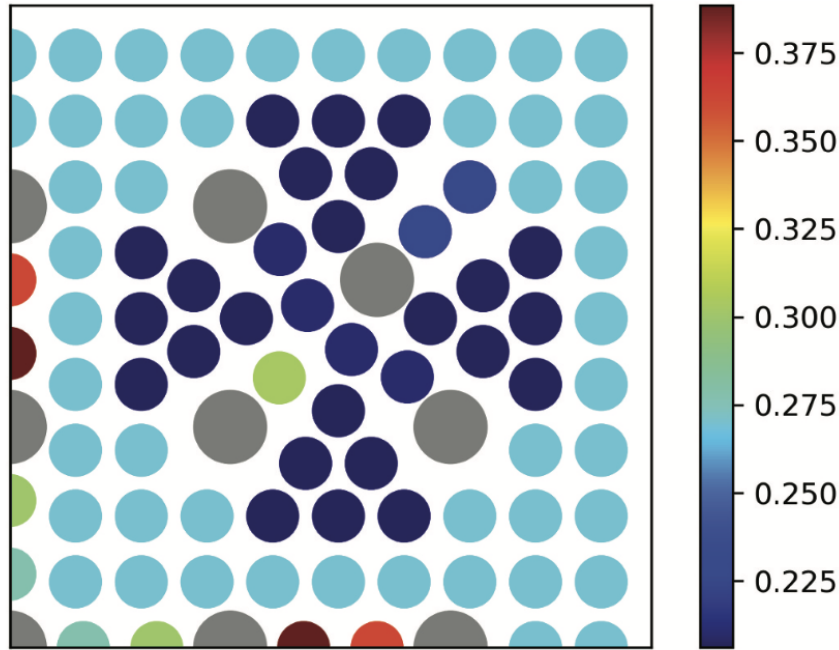


Figure 1: The geometry of the BL324 assembly, from [10]. The color represents the distance of pins from their nearest neighbour (excluding guide tubes) in centimetres.

The design aimed to bring fuel pins as close as possible to each other while respecting the minimum spacing. The outer two rows are evenly spaced pins, with triangular close packing between the guide tubes. Other pins are placed on the diagonal of the assembly and the remaining pins are placed "by hand" to maximise the gap with adjacent pins. We can indeed see in figure 1 that many of the pins are as close to each other as possible within the requirement of a 2 mm gap. This was seen by Lindley as a sign that the design was close to optimal.

A uniform 4.95 wt% enrichment is used at the beginning, it is the current enrichment limit for PWR fuel. A simulation was also run with a 5.95% enrichment to model a power uprate of 20%. This is an enrichment in the range of low enriched uranium plus (LEU+) fuel, between 5% and 10%. This range of enrichment is currently being studied by manufacturers [11] [12]. This higher enrichment is not needed to achieve a power uprate if one was to accept a different fuel management strategy, with shorter cycles or lower burnups for example.

Power peaking BL324 was designated as the most successful new assembly since it achieved a high number of pins and a low power peaking. Indeed, it had a power peaking of 1.11 within

one assembly, only 2.1% higher than a traditional assembly. Since BL324 contained 22.7% more pins than this reference, in principle the assembly power could be increased by $22 - 2 = 20\%$ while maintaining peak power temperature.

Lindley notes that this result can be improved if the enrichment of the hottest pins is lowered, which will reduce power peaking. In a test with enrichments of 4.95% and 4.75% in different zones the power peaking factor was limited at 1.08. A similar result was achieved for 5.95 and 5.75%, which yielded a power peaking factor of 1.09.

Other requirements The control rod worth did not vary much with the number of pins, but a higher enrichment did degrade it.

The evolution of reactivity with burnup was also correct with the BL324 assembly when the cladding thickness was proportional to pin size. The value of k_{∞} was within 300 pcm of the reference assembly for all burnup values. This is not surprising since the amounts of fissile material, moderator and cladding were all kept the same.

When the cladding thickness is kept constant Lindley sees that the value of k_{∞} decreases faster than BL324 with proportional clad or the reference. In this case, the cycle length was reduced by 3.5%, the same variation as the reduction of the fuel volume due to the increased cladding. This negative consequence was expected but needs to be taken into account for applications, since keeping a constant cladding thickness might be needed due to practicalities in manufacturing or safety requirements.

The pressure drop of BL324 was 103.6 kPa compared to 100.6 kPa for the reference assembly. Lindley considers that this change is acceptably small. But he stresses the importance of modelling transition cores with both new and old assemblies to actually see if this pressure difference has an impact.

The MDNBR is even better for BL324 rather than the reference assembly, even without enrichment zoning. Lindley explains this because the hottest pins are not in the same places as the smallest channels for water to flow, see figure 2.

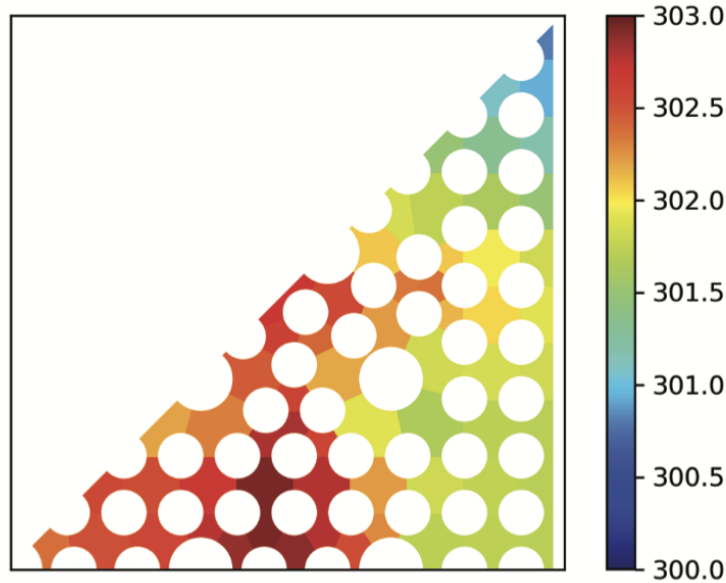


Figure 2: Subchannel outlet temperatures ($^{\circ}\text{C}$) of the BL324 assembly, from [10].

3

PROJECT METHODOLOGY

In this project, formal optimization has been applied to the problem described above : increasing the number of fuel pins per assembly to allow for power uprates in current PWRs.

We studied the case of 324 fuel pins, the number that Lindley [10] has proposed. While his design was based on his intuition of the best way to arrange fuel pins, we have tried to use an optimization loop to determine the best positions for all the pins.

For an optimization to run, an objective or objectives (figures of merit) are to be maximized or minimized. Here, different choices can be made. Based on Lindley's reasoning, we considered in a first part that the goal of allowing a power uprate while maintaining peak fuel temperature is determined by the number of fuel pins and within-assembly power peaking. For this reason, power peaking is a good choice of optimization parameter, since it has a direct impact on the possible power uprate. To take into account the MDNBR, in this first part we consider it as a constraint : proposed optimal assemblies with an MDNBR lower than a traditional assembly will not be accepted.

In a second part we recognized that the MDNBR could actually be a limiting factor for the power of current PWR assemblies, rather than the maximum fuel temperature. This was thanks to a remark made by Eugene Swageraus. Considering that the maximum fuel temperature was what limited the current PWR power led us to consider the power peaking as a variable to improve. But instead we have tried in a second study to allow the power peaking to be slightly

worse while improving the MDNBR, which could ultimately allow us to have a power uprate above 20% while maintaining all safety parameters.

3.1 MODELLING OF ASSEMBLIES : SERPENT AND COBRA

The modelling of the assemblies was done using Serpent [13] for neutronics and COBRA [14] for thermo-hydraulics.

The choice of COBRA comes from the fact that Lindley used it in his paper [10] and gave us access to his input files, as well as the fact that this software has been used previously in NE2.

The choice of Serpent comes from the knowledge and availability of this software in the Cambridge University nuclear research group. We could have used OpenMC [15] like Lindley and used his input files like for COBRA. However, this presented the risk of not finding help among Cambridge researchers for software-specific issues we might run into. The project supervisor also emphasized the importance of writing an input file alone to familiarize oneself with the code and the original assembly design.

Using SCONE [16] was also possible, but it is a code designed for research on the Monte Carlo method itself rather than its applications. Serpent can be more optimized for modelling a PWR assembly.

In a last study WIMS [17] replaced Serpent for neutronics simulations. Indeed, since Serpent is a Monte Carlo method, its results are stochastic and presented large uncertainties that made optimization difficult.

3.2 OPTIMIZATION LOOP

For a fixed number of pins, their size is first determined from the conservation of H/HM ratio. We consider that all pins have the same size, based on the requirement that the heating of coolant is even across the assembly.

Then the position of each pin can be varied continuously within a given area based on limiting parameters such as the minimum distance between two pins and the total size of the assembly. The optimization begins from a given set of pin positions, which was the one selected by Lindley. Other possible designs can then be explored, and the algorithm will follow a path towards the lowest value of power peaking.

The limits on our possible assemblies are also taken into account by the optimization process. If two fuel pins are too close to each other or if a pin and guide tube are overlapping the design is rejected before the Serpent and COBRA models run. The optimization also needs to

take into account the limit on MDNBR, which is obtained from the COBRA calculation. We chose to simply reject the proposed assemblies that fall outside this limit. This option was easy to adopt, since the MDNBR of the new proposed assemblies are almost always higher than the reference (traditional) assembly and thus pose no problem. A design with an MDNBR lower than the traditional assembly would be rejected.

We end up with the loop show in figure 3 : the Python optimizer calls Serpent for the neutronics calculations, which yields the power peaking. Serpent then passes the pin power list to COBRA, which runs a thermohydraulics computation to calculate the MDNBR. Both these values are handed back to Python for the optimizer to generate a new design candidate.

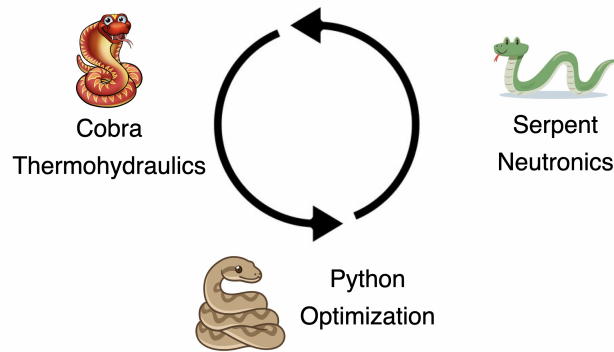


Figure 3: The optimization loop with Serpent, COBRA and Python.

4 SIMULATIONS SETUP

4.1 SERPENT

4.1.1 • FIRST MODEL

The first work done was to set up a model of BL324, Lindley's proposed assembly for a power uprate.

We normalize the fuel assembly power to 17.7 MW as Lindley states in his COBRA parameters. This is a traditional PWR power, not an uprate scenario. We also specify temperatures of 900 K for the fuel and 600 K for the coolant for the cross sections.

Full assembly geometries First a model of a reference 17x17 traditional PWR assembly was created. This was done using the usual Serpent method of pin universes, then a lattice,

and an assembly nest.

The detector for each pin's power is a mesh detector with size $L_{assembly}/17$. We eliminate the guide tubes from this 17x17 result when we analyze it, in order to calculate power peaking among the fuel pins.

The BL324 assembly could not be modelled using this method since it does not follow a regular lattice. Instead the geometry was described using the CSV definition, simply creating cylindrical surfaces for fuel, gap and cladding centred on each pin position. This of course could not be done by hand since there are 324 pins plus the usual guide tubes.

The position of each pin was taken from Lindley's paper [10] and symmetries were used to go from this 1/8th of an assembly list to the positions of all the 324 pins. A Python algorithm generates the definitions of surfaces and cells needed to create the geometry. The assembly surface is then added. A last cell is created for coolant inside the assembly surface and outside the cladding surfaces of each pin.

The detectors for pin powers are based on cells, with one detector per fuel cell.

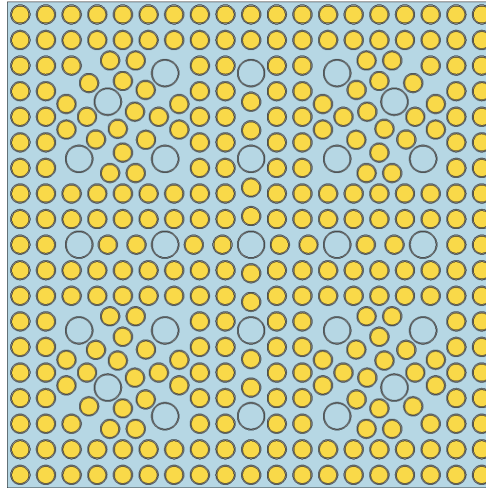


Figure 4: The geometry of a BL324 assembly as given by the Serpent output. In blue is the water, in gray cladding and in yellow the fuel.

We get for the reference assembly a power peaking factor of 1.077. We can see in figure 5 which pins are the hottest. This is similar to what Lindley finds : the reference design is his case A, he quotes 1.09 as the power peaking for this case.

For the BL324 assembly we get power peaking = 1.120. We can see in figure 6 that the hottest pins are the ones closest to the center of the assembly and next to the guide tubes, much like in the reference assembly.

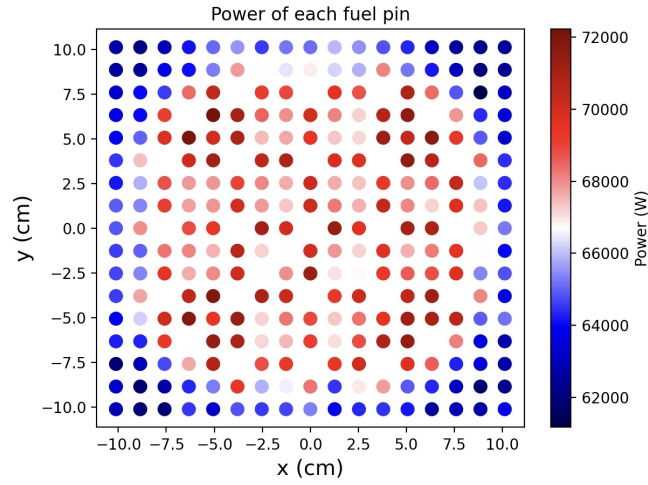


Figure 5: The pin powers for the reference assembly.

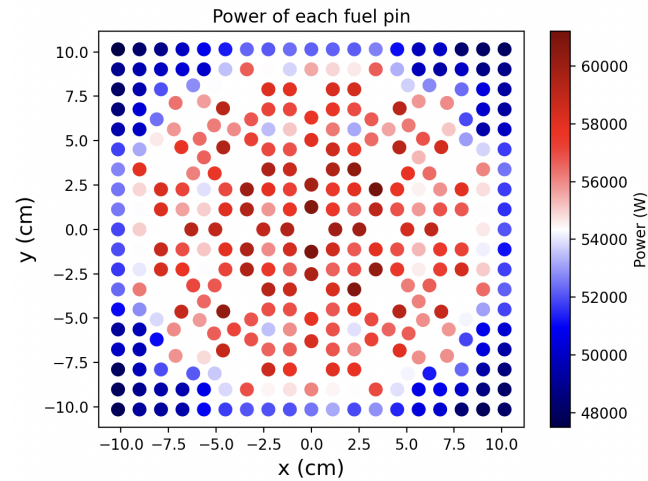


Figure 6: The pin powers for the BL324 assembly.

We can compare our BL324 result with Lindley's result shown in figure 7. The hottest pins are in the same positions, but Lindley gets a 1.11 power peaking factor, which is slightly lower than ours.

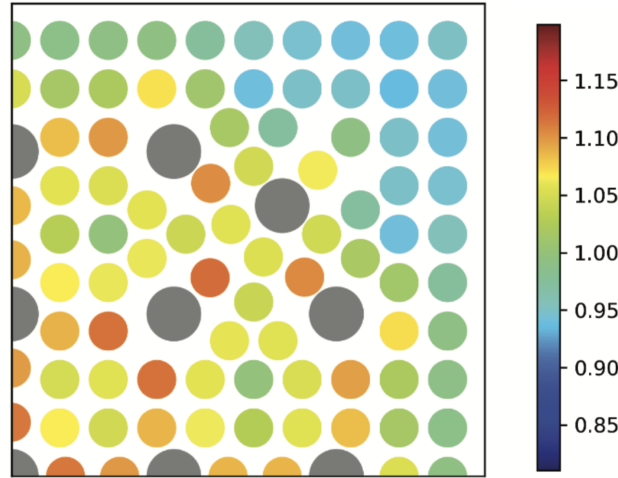


Figure 7: Pin power relative to the average power, from [10].

4.1.2 • SYMMETRIES OF THE ASSEMBLY

To limit the number of histories needed to get an accurate result, we use the symmetries of the full assembly and only model on part of it. First, we use the quarter symmetry. We simply plot only the pins in this quarter ($x > 0$ and $y > 0$) and implement reflective boundary conditions at all boundaries.

As we can see in figure 8, the pins in the inside edges are only half represented since they intersect the surface of the quarter assembly. In order to calculate pin power the result of the Serpent tally must be multiplied by 2 for these edge pins. The pin situated at (0,0) is a guide tube so it doesn't have to be considered uniquely.

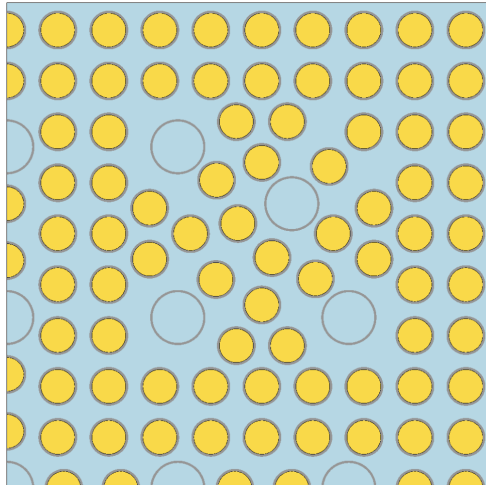
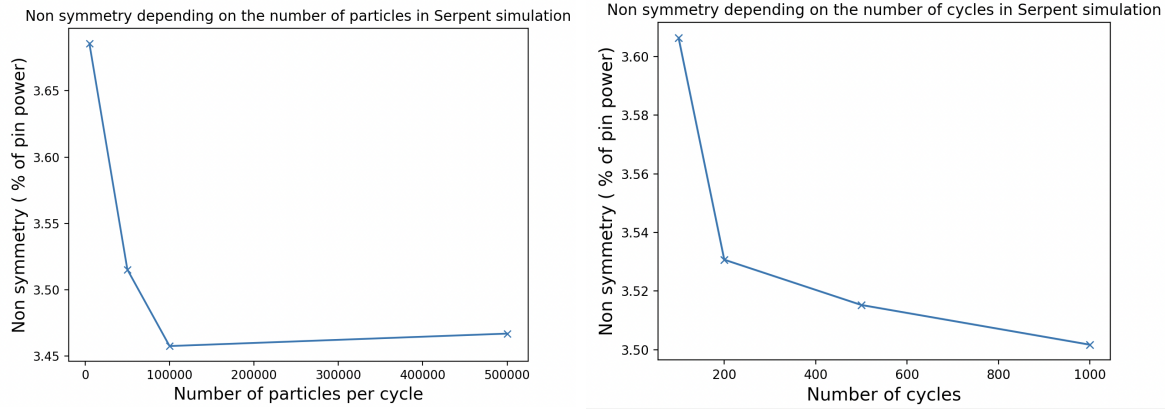


Figure 8: Geometry for the simulation of 1/4th of a BL324 assembly.

Models of the quarter assembly were used to gain insight into the convergence of the sim-

ulation. There are two sets of information that we need from the Serpent simulation for the optimization process : power peaking, the figure of merit in our optimization process, and the power of each pin, which allows us to compute the MDNBR for the assembly. While monitoring the convergence of k_{∞} is often enough to conclude on the suitability of the number of histories for a Serpent calculation, here we wanted to have a better measure of the reliability of the results that we actually need. One indicator used was the "non-symmetry" of the pin power : the average percentage of power difference between symmetric pins, which are supposed to have the same power. We can see in figure 9b and 9a that the non-symmetry quickly decreases with the number of histories. However it does not go down to 0, instead we have a minimum non-symmetry of about 3.5%. This is not surprising, due to the stochastic nature of the Monte Carlo simulation. So this value of error will be used for the individual pin powers in the future, rather than the stochastic uncertainty on k_{∞} which is much lower. However, the way this error propagates into the power peaking value is not clear, and it is even less clear for the MDNBR value after the COBRA simulation. The errors on MDNBR and power peaking will be evaluated through a series of direct tests by running Serpent and COBRA a high number of times and plotting their results compared to a normal distribution.

We can compare this "non-symmetry convergence" to the traditional measure of convergence k_{eff} in figures 10b and 10a. The evolution of this indicator is less monotonic because by the number of histories chosen k_{eff} has already converged well; the large number of particles and cycles are only needed to resolve more finely the individual pin powers.



(a) Non-symmetry after 500 cycles depending (b) Non-symmetry depending on the num-
on the number of particles in the Serpent sim- ber of cycles in the Serpent simulation (with
ulation 50,000 particles per cycle)

Figure 9: Non-symmetry convergence

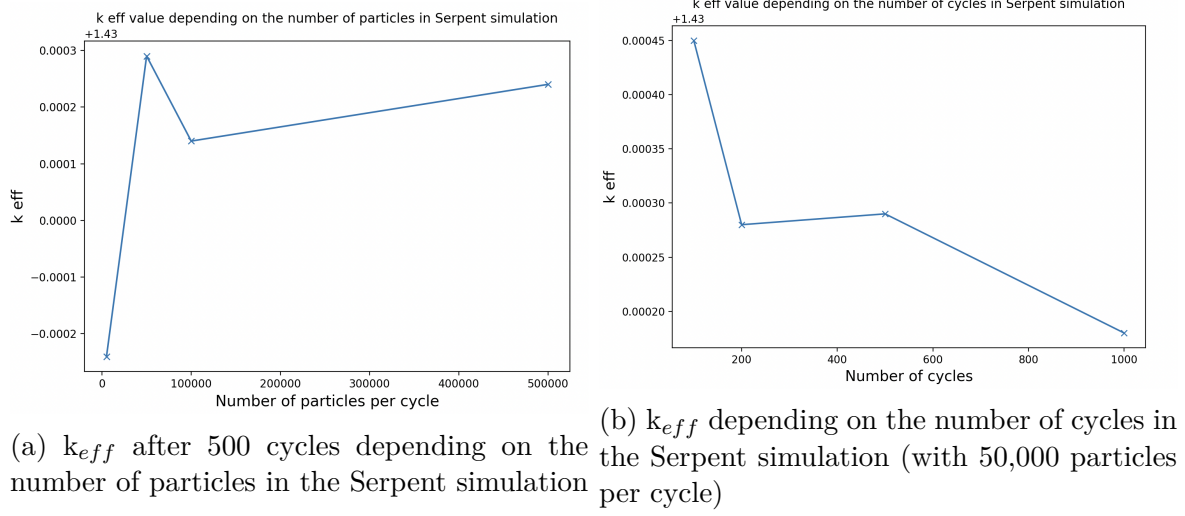


Figure 10: k_{eff} convergence

Final model The full symmetry of the assembly is 1/8th, with a line of symmetry along the diagonal of the quarter assembly. This is implemented in Serpent by setting up a universe symmetry for the universe that contains every element of our assembly (pins, coolant, etc.). The geometry plotted is the same but we only need to input the position of the pins in the lower half of the quarter assembly ($x > y$). We need fewer histories for a similar quality of result.

In this situation we also need to count twice the power of the pins on the diagonal since only half of their surface has actually been modelled. We get a result of power peaking of the form seen in figure 11. The value of power peaking is 1.09, which is lower than Lindley's result. We have checked this result against the full assembly model, both for power peaking value, position of hot and cold pins and k_{∞} to confirm that the use of symmetry does not change the results.

4.1.3 • COMPARING DIFFERENT SIZE OF MODELS

We have modelled in Serpent the same BL324 assembly as a full assembly, a quarter and 1/8th. While in theory these three models should give us the same results, and we do get the same values of k_{∞} we see differences in the values of power peaking. We can see these values in table 1 below.

Serpent	Full assembly	1/4th	1/8th
Power peaking	1.12	1.10	1.09

Table 1: Power peaking factors for the Serpent model at different assembly fractions.

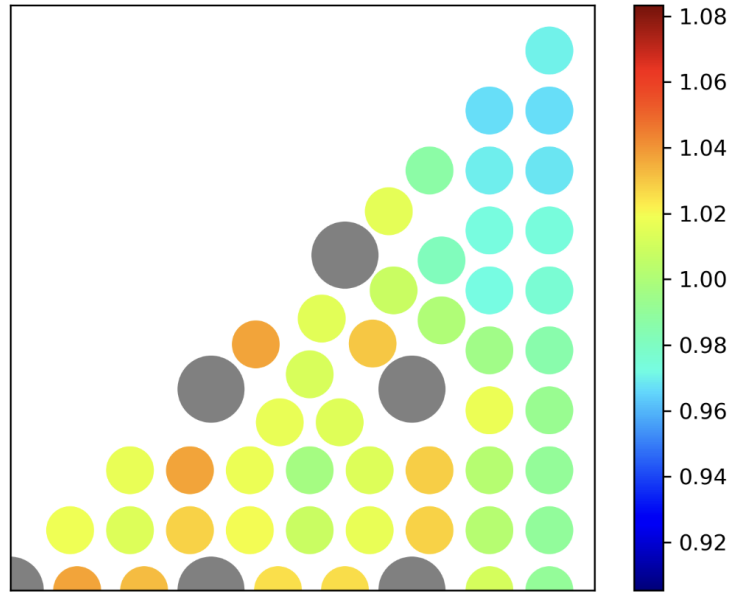


Figure 11: Power for each pin in the 1/8th of assembly modelisation.

It seems that the power peaking is higher when we model a larger fraction of the assembly. Indeed, as we have seen above when studying non-symmetry in a quarter assembly, the powers of symmetric pins are not exactly equal. If we have 8 examples of the hottest pin instead of one, we have more chance for one of those pins to have a slightly higher power. At the same time, the average pin power does not change, since it is controlled by a normalization. This inherent uncertainty of the Monte Carlo method means that increasing the fraction of the assembly modelled increases the power peaking.

Since Ben Lindley models a quarter assembly in OpenMC and gets a peaking factor of 1.11 we can conclude that our model is accurate and corresponds to Lindley's work. This is true even if the values obtained for 1/8th of an assembly that we will work with in the rest of this project will be lower than Lindley's power peaking.

4.2 COBRA

For the COBRA simulation the code written by Ben Lindley was mainly used. He had already set up a simulation of an irregular assembly in COBRA and his parameters were kept the same as much as possible.

We are modelling the assembly as an octant, with an active fuel length 3.66 m, divided into 16 equal axial nodes. Radial power distribution is taken directly from the Serpent calculation and normalised to a total assembly power of 17.7 MW, with quadrant symmetry giving around 4.4 MW to the modelled octant. Axial power shape is modelled as a chopped cosine with peak-to-average ratio of 1.55. We also have 8 grid spacers at uniform axial positions, with a

loss coefficient of 1. Other parameters can be found in the table 2 and represent average PWR operating parameters.

Models used are from the Electric Power Research Institute (two-phase friction, subcooled boiling, void fraction). We use the W-3 correlation for critical heat flux (CHF).

System pressure	15.5 MPa
Inlet coolant temperature	292°C
Coolant temperature rise	30°C
Inlet mass flux	2804 kg/m ² s
Turbulent mixing coefficient	0.038
Power fraction in coolant	2.6%
Cross-flow resistance	0.5
Transverse momentum factor	0.5

Table 2: Values of parameters for the COBRA simulation

COBRA solves the three conservation equations (mass, energy, axial momentum) for each subchannel simultaneously, with lateral crossflow exchange between adjacent subchannels governed by the diversion crossflow and turbulent mixing models. At each axial node and each rod, the local heat flux q'' is computed from the power of the pin and the heated perimeter. The W-3 critical heat flux correlation then predicts the critical heat flux q''_{CHF} as a function of local pressure, mass flux, quality, and hydraulic diameter. The departure from nuclear boiling ratio at each location is $DNBR = q''_{CHF}/q''$. The MDNBR is given as the minimum DNBR across all subchannels and axial positions : COBRA evaluates this at each axial node for each rod and reports the global minimum.

When computing the MDNBR we have to take into account more departure from nuclear boiling-prone conditions such as operational transients as well as uncertainties. We kept Lindley's conditions as shown in table 3 although more negative conditions could be assumed. With these conditions Lindley finds a 1.91 MDNBR in the traditional PWR assembly.

We did have to make some modifications to Lindley's original code. First we had to ensure that the Serpent output was correctly linked to provide the right linear power of each pin to the COBRA simulation. Then there was the larger problem of defining subchannels.

Flow rate modifier	95%
Inlet temperature modifier	+2°C
Core power modifier	112%
Radial power peaking factor	1.587

Table 3: COBRA parameters for MDNBR study

Subchannels Subchannels are the blocks of coolant between a group of pins, connected with small ‘gaps’ to the neighbouring subchannels. Lindley’s approach was to print out the design of the assembly, number the pins as they are in the original pins list, decide heuristically where subchannels should be and define them by the number of the pins that create the subchannel. He then wrote a code to create the required mappings to input in COBRA from this list of pin indexes. The guide tubes are treated separately and noted in the subchannels list by negative indexes. These subchannels can be seen in figure 12a.

However, in our case where the pins are moving, defining subchannels has to be done automatically. For a simpler solution we started by assuming that every subchannel will be a triangle between three closest pins. This is akin to defining a mesh. With the help of Nathan Ravoisin the method of Delaunay triangulation [18] was tried, which did provide a correct set of "triangle subchannels". It is implemented in Python in the scipy library as `scipy.spatial.Delaunay` [19]. The edge subchannels were added in manually. The resulting subchannels are shown in figure 12b.

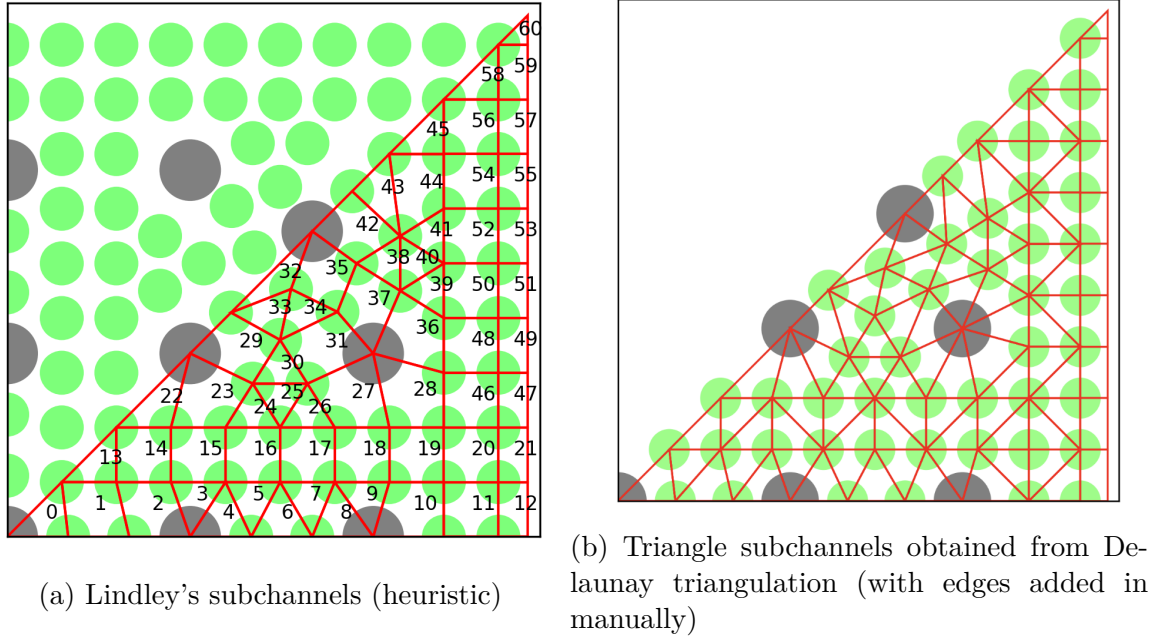


Figure 12: Different choices of subchannels

The numbering of the pins can be seen in figure 13. It is important that this does not change since the edge subchannels were added in manually based on this numbering of the pins and, as we will see in the following sections, pins are referred to by their index when moving them sequentially. Identifying which pins are on the diagonal is particularly crucial to being able to move these pins using only one coordinate. This numbering does not correspond to Lindley's for reasons of logic (having pins close together share similar numbers is more practical).

There is a difference of MDNBR result between both sets of subchannels. It seems that the triangular subchannels lead to higher (more favourable) MDNBRs. Indeed, as an example, one run of Serpent on the original BL324 design gave for Lindley's channels a MDNBR of 1.943 while for our channels we had a MDNBR of 1.965. Since these two COBRA runs followed a single Serpent run they were given the same pin powers, so COBRA itself gives different MDNBR values for different subchannel setups.

The difference between the two subchannel designs cannot at the moment be attributed to one design being better than the other. It was decided to consider this difference as an uncertainty on the values of MDNBR that comes from the COBRA setup. However this is not much of a concern for now since a traditional PWR assembly has an MDNBR around 1.91, so our design is safer than the traditional assembly and the MDNBR does not need to be precisely resolved. It is also important to note that both of these MDNBR values come from setting up COBRA parameters through approximation and using critical flux correlations which are not certain either.

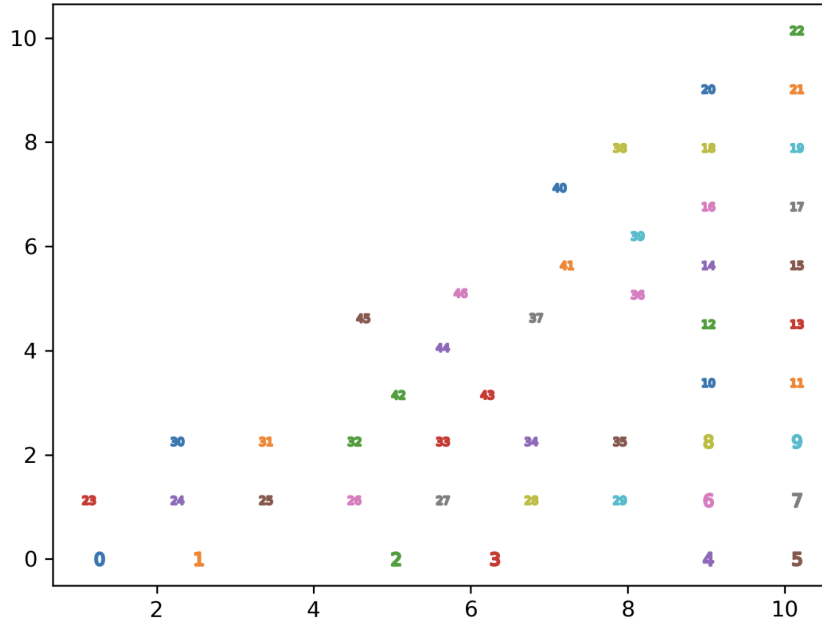


Figure 13: Indices of the pins in the new pins list for the original BL324 assembly

5 THE FIRST OPTIMIZATION

5.1 THE PROCESS

The workflow of the optimization process is as follows :

- The optimization code proposes a set of pin positions,
- The objective function is called to evaluate the power peaking and MDNBR,
- First, we check that the geometry is allowed. If a pin is too close to another pin or overlaps with a guide tube, the objective function immediately returns a figure of merit of $F=100$. The rest of the loop does not happen. If not, we continue :
- The code `BL324 1/8 maker.py` generates the Serpent input code `BL324 1/8`,
- Serpent runs with a `subprocess.run` command from the objective function,
- The `calc P` function gets the Serpent results from the `BL324 1/8 det0.m` output file to get the power of each pin,

- The `heat map` function uses this pin power list to calculate the power peaking and to draw the map of the assembly with the relative power of each pin. The figure of merit F is the power peaking,
- Using the pin powers a COBRA input is generated by the function `cobra input`,
- COBRA is run with a `subprocess.run` command, with worse conditions than the average assembly to get the real MDNBR,
- The function `cobra output` reads from the output file `OUTFILE` to get the MDNBR and the pressure drop of the assembly,
- If the MDNBR is too low, the figure of merit is set at $F=100$,
- End of the objective function, the figure of merit is returned to the optimizer. The optimizer can generate a new set of pin positions.

5.2 THE ALGORITHM

Many algorithms traditionally used (such as `scipy.optimize`) do not work with a noisy figure of merit function. Indeed, we do not have

$$x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$$

which can cause many calculations of gradients to break down. A method like `scipy.optimize` [20] would have been a good solution for our problem since it allows constraints, such as non overlap of pins, to be handled separately from the objective function.

We looked for an optimization method made for an objective function that is both noisy and long to run (considering the complexity of a Serpent run). We chose the `skopt` (scikit-optimize) functions, especially `gbdt minimize` [21]. This method relies on gradient boosted regression trees. First the usually expensive to evaluate objective function is approximated by a surrogate model [22]. The concept of such a model is to create an approximation of the optimization parameter (here power peaking) as a function of pin positions. Then the optimizer will use this model to determine a potential minimum. We will evaluate the expensive function at the next best point and this evaluation will be used to both improve the surrogate model and determine if this point is indeed the minimum. This allows us to find the minimum with as few evaluations using Serpent as possible.

The optimizer allows us to give a maximum number of calls to the objective function, to have a somewhat fixed runtime. However, for our case, it will be more of a maximum since when the proposed position leads to overlap between pins Serpent is not called and the objective function is very quick to run (compared to Serpent at least).

5.3 THE SERPENT FIDELITY PROBLEM

In order to run the optimization, we need to have a good certainty for the power peaking value. In particular, the uncertainty on the power peaking needs to be lower than the difference caused by moving a pin, for the optimization to work.

Uncertainty of Serpent First, we have studied the uncertainty on the result of Serpent and COBRA : power peaking and MDNBR values. This uncertainty cannot be easily deduced from another value like the non-symmetry on pin power (since the power peaking relies on a maximum function and the MDNBR calculation is even more complex). So we studied the stochastic effects directly by running sets of 100 runs of Serpent with the same number of particles per cycle and cycles.

Such sets of runs fit a normal distribution quite well. It is important to note that we are only running 100 Serpent +COBRA cycles and sorting the results in 25 bins, which explains the lack of smoothness of the plot. We can see in figures 14b, 14a, 15a and 15b the plot of the number of Serpent results in each interval of power peaking or MDNBR values, for different numbers of histories. The uncertainty does get lower as we increase the number of histories.

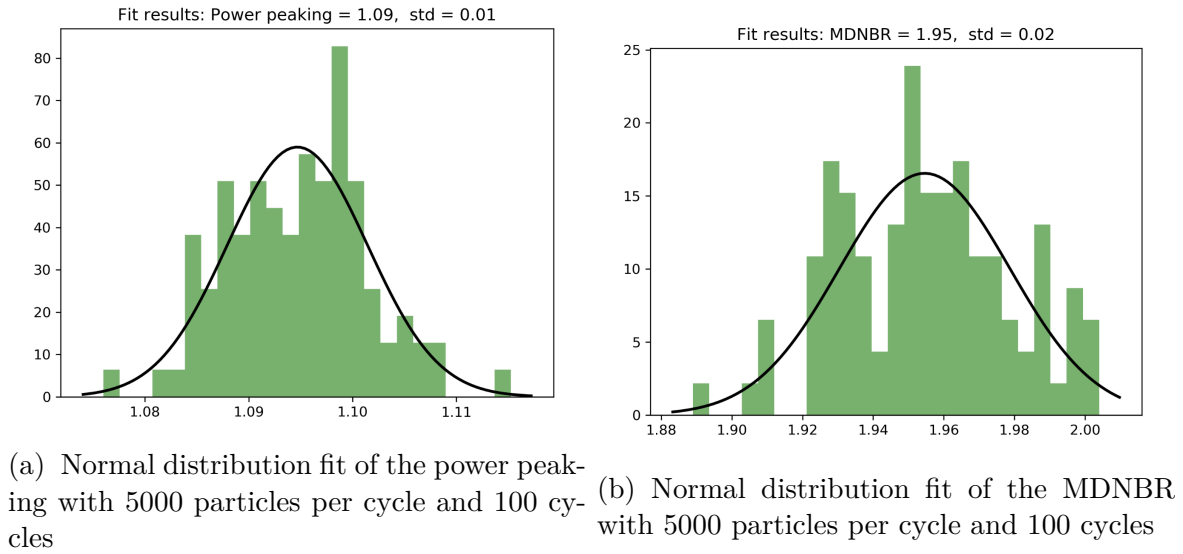
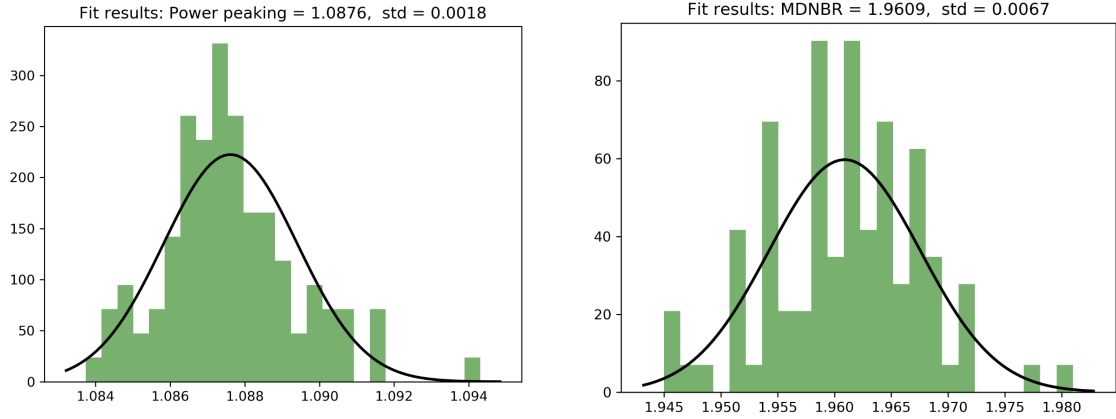


Figure 14: Normal distributions of results for low precision Serpent evaluations

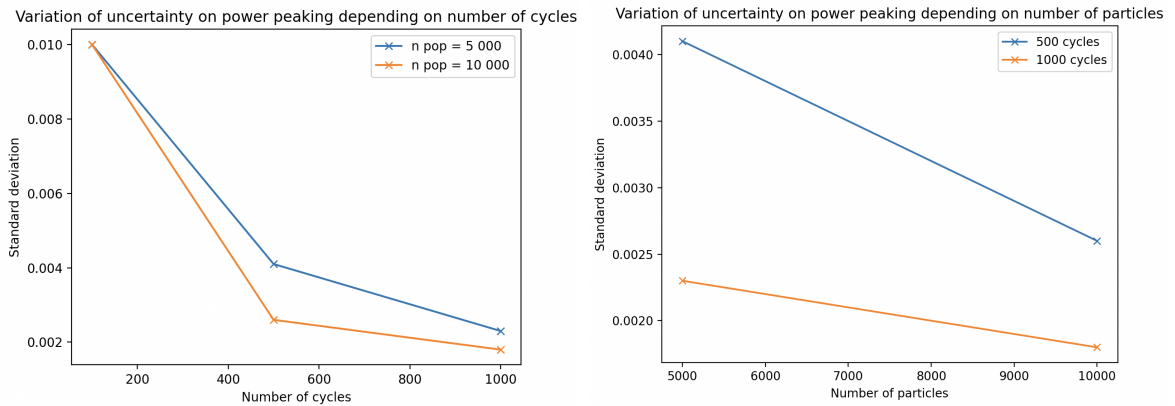


(a) Normal distribution fit of the power peaking with 10000 particles per cycle and 1000 with 10000 particles per cycle and 1000 cycles (b) Normal distribution fit of the MDNBR with 10000 particles per cycle and 1000 with 10000 particles per cycle and 1000 cycles

Figure 15: Normal distributions of results for high precision Serpent evaluations

In general, the number of histories is the main factor on which the precision of the result depends, as seen in figure 17. The fact that the values of uncertainties for the same number of histories but a different number of cycles and particles per cycle is reassuring : the value of the standard deviation calculated makes sense and is not itself subject to too much uncertainty.

We can compare the standard deviation for different numbers of particles or cycles better in figures 16a and 16b.



(a) Standard deviation of the power peaking depending on the number of cycles (b) Standard deviation of the power peaking depending on the number of particles per cycle

Figure 16: Evolutions of standard deviations

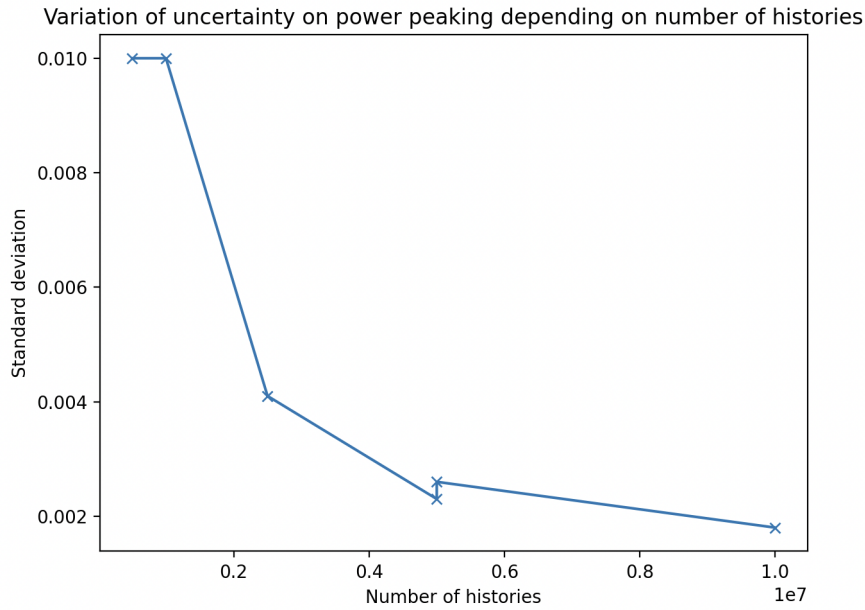


Figure 17: Standard deviation of the power peaking depending on the total number of histories

The need for precision We tried to estimate the need for precision in the optimization by estimating the variations of power peaking when we move one pin. We have chosen to study pin 45, which is easy to move since it has a large space available before hitting another pin, as well as being a hot pin whose position has an effect on the power peaking, see figure 18.

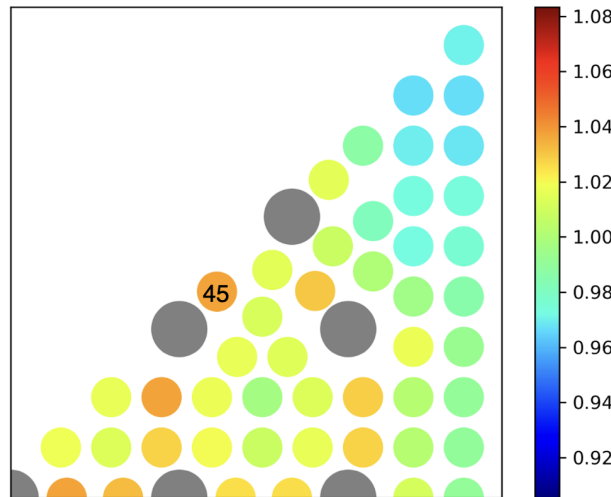


Figure 18: The position of pin 45 in the BL324 assembly. The colormap represents power peaking.

We are moving pin 45 along the diagonal of the assembly, between the position where it is touching the guide tube and the one where it hits the minimum distance between pins with

pin 46 and its symmetric twin. We use 5000 particles per cycle and 1000 cycles. This means that estimations take a long time, about 150 s for each calculation of the figure of merit (non overlap check + Serpent + COBRA).

We can see this study in figures 19 and 20. The uncertainty has to be much lower than the variation of power peaking when moving one pin in order for the optimization to be meaningful. Here, we are using one standard deviation for the error bar of the results.

It seems that the MDNBR does not vary much when this pin is moved, perhaps because it is not a limiting pin. However pin 45 is indeed an important pin for the power peaking, which does vary.

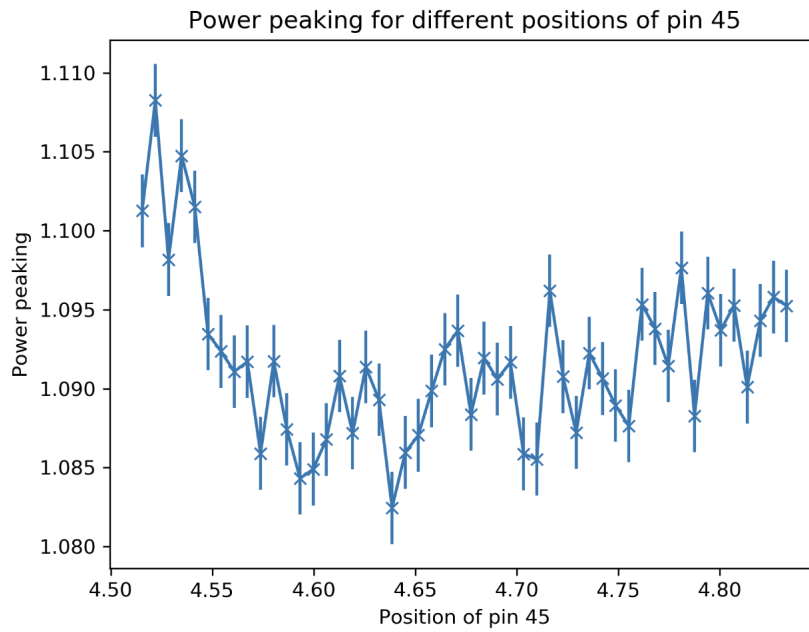


Figure 19: Power peaking for each allowed position of pin 45 on the diagonal of the assembly.

It seems that having pin 45 far from the guide tube reduces its power since it is less moderated, as we can see in figure 21a compared to figure 21b. Since pin 45 was originally a very hot pin we have a reduction of the power peaking when we reduce the power of pin 45. One may note that not all plots shown here are coherent with this observation. This is indeed the case since Serpent uncertainties are so high that, quite often, the general trend will be entirely unrecognisable.

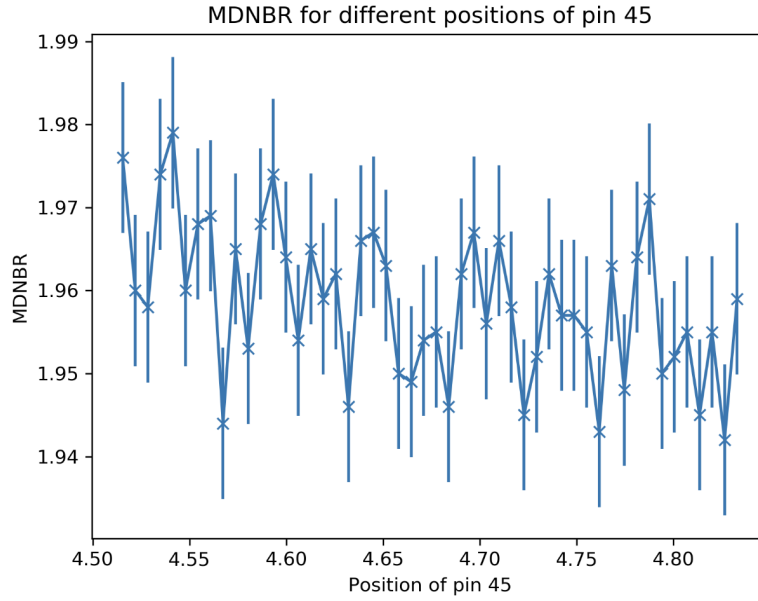
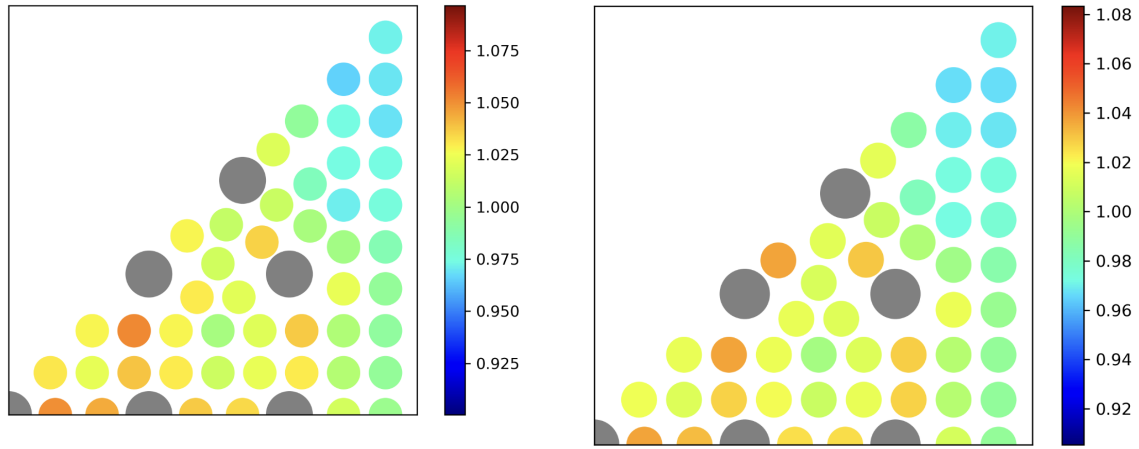


Figure 20: MDNBR for each allowed position of pin 45 on the diagonal of the assembly.



(a) Power of each pin with pin 45 moved as close to pin 46 as possible, or far from the (b) Power of each pin with the usual BL324 guide tube. assembly.

Figure 21: Power of each pin for different positions of pin 45

Even with the maximum number of histories studied (10000 particles, 1000 cycles) we have for 100 runs of Serpent a variation of 0.01 between results (the extreme values are 1.084 and 1.094), as seen in figure 15b. We have studied variation of power peaking over the full possible movement of a pin that influences the power peaking a lot (pin 45, on the diagonal) in figure 19, the amplitude of variation is about 0.03. This is why using Serpent to perform an optimization will be very difficult.

Indeed we can see that the solution presented in figure 22 is not at an optimum. We ran a set of Serpent evaluations during an optimization which gave us this optimal point. During the Serpent run in this optimization process the "optimal" position had a lower power peaking than the original. Then we looked at the power peaking for both positions using values from another set of runs of Serpent (the one shown in figure 19). As we can see, for this set, the "optimal" position has a power peaking equal to that of the original position. Indeed, each run of Serpent is liable to giving a new "optimal" position, which means that the optimization is meaningless.

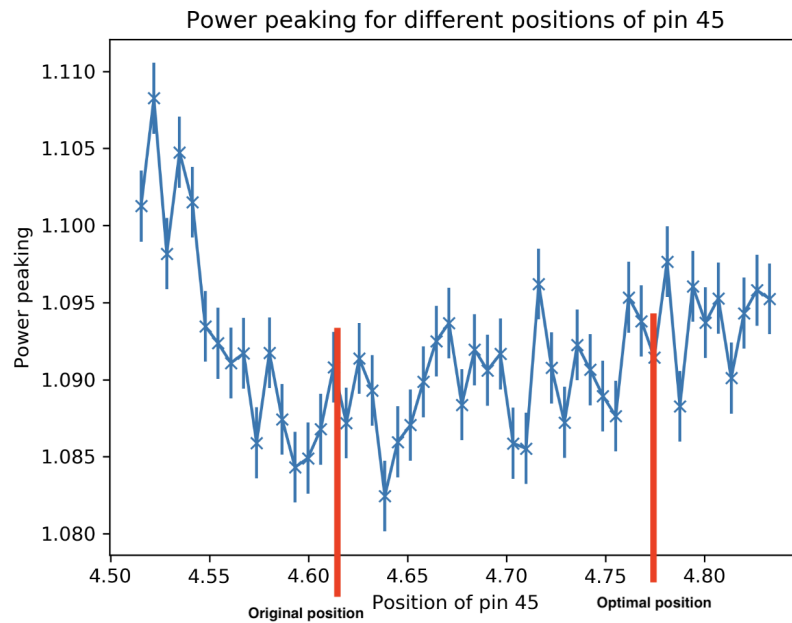


Figure 22: The position of pin 45 in the original assembly and as a result of the optimization

6

OPTIMIZING FOR THE MDNBR

In a second part we realized that the MDNBR could actually be a limiting factor for the power of current PWR assemblies, rather than the maximum fuel temperature. Considering that the maximum fuel temperature was what limited the current PWR power led us to identify the power peaking as a variable to improve. But instead we can allow the power peaking to be slightly worse while improving the MDNBR, which could ultimately allow us to have a power uprate above 20% while maintaining all safety parameters.

The idea was to use a single run of Serpent, which could for this reason be a very expensive, very detailed run, and hypothesize that the pin powers would not vary much as we moved pins around. This would allow us to run an optimization using only COBRA runs, which was computationally cheaper and did not have issues of uncertainties. Thanks to this deterministic behaviour we used `scipy.optimize.minimize` [20], the standard Python optimizer. This method only allows deterministic objective functions which can be seen as continuous. Since we were trying to maximize the MDNBR rather than minimize it, we used $F = -\text{MDNBR}$ as an objective function and set $F = 0$ in cases where we had an overlap between pins.

We allowed the optimizer to move every pin except the ones on the edge of the assembly (right hand column), which needed to stay in place to define the edge subchannels, and the bottom row of our 1/8 assembly which already seemed well organized. The pins on the diagonal were only allowed to move in one dimension (move along the diagonal but remain on it, with $x = y$) and all the other pins had full 2D freedom of movement.

The optimizer did succeed in finding a better set of pin positions, which we can see in the evolution plotted in figure 23. The large run of Serpent used for the original pin power values had a population of 500000 particles per cycle and 1000 cycles, which is the highest precision we have used in this project. The parameters given to the optimizer included to stop when position changes were under 0.1 mm or when the figure of merit changed by less than 0.01 because the MDNBR is only given with 3 significant figures by COBRA. The optimization converged in about 300 calls to COBRA, although more designs were proposed which resulted in overlapping geometries and were immediately rejected.

We compared results between the MDNBR-optimized assembly and the original BL324 design. To fully study this new configuration we did 100 runs of Serpent with 10000 particles per cycle and 1000 cycles. Just like in figures 15a and 15b we plotted the normal distribution fit in figures 24a and 24b. For the original BL324 we had a MDNBR of 1.9609, with a standard deviation of 0.0067. For the optimized assembly we see a MDNBR of 1.9882, with a similar standard deviation of 0.0059. This is indeed a clear improvement, although it is less stark than the COBRA runs based on the original pin powers used during the optimization showed. Indeed, the optimizer believed that the MDNBR of the optimized assembly was above 2.

The power peaking is mostly stable, which is expected since our pin powers have been proven

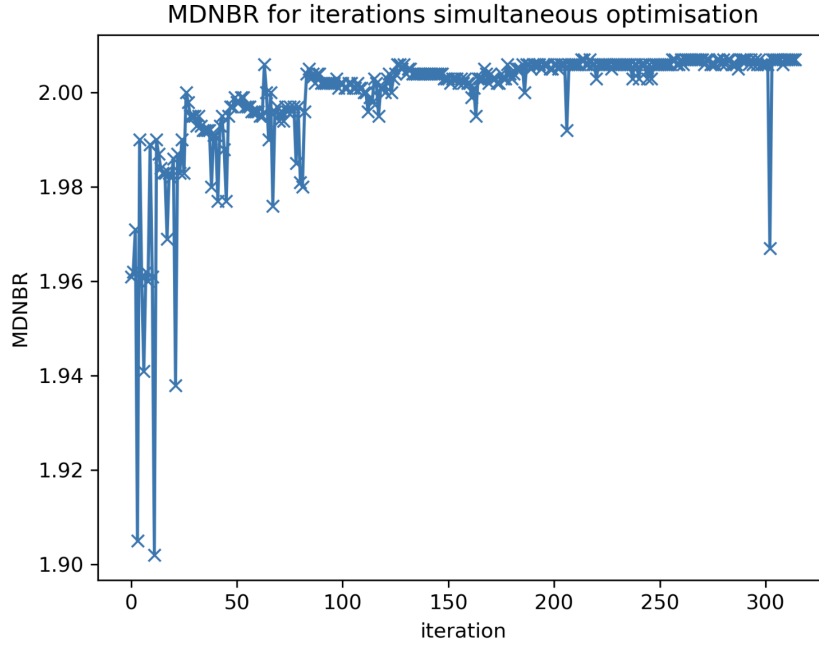


Figure 23: The MDNBR for each set of pin positions studied by the optimizer

to be stable as well. Still, the power peaking of the MDNBR-optimized assembly is higher than the original, with a difference of 0.002 which corresponds only to a 0.184% increase. The original BL324 power peaking is 1.0876 with a standard deviation of 0.0018 while the optimized assembly has a power peaking of 1.0896 with a standard deviation of 0.0024.

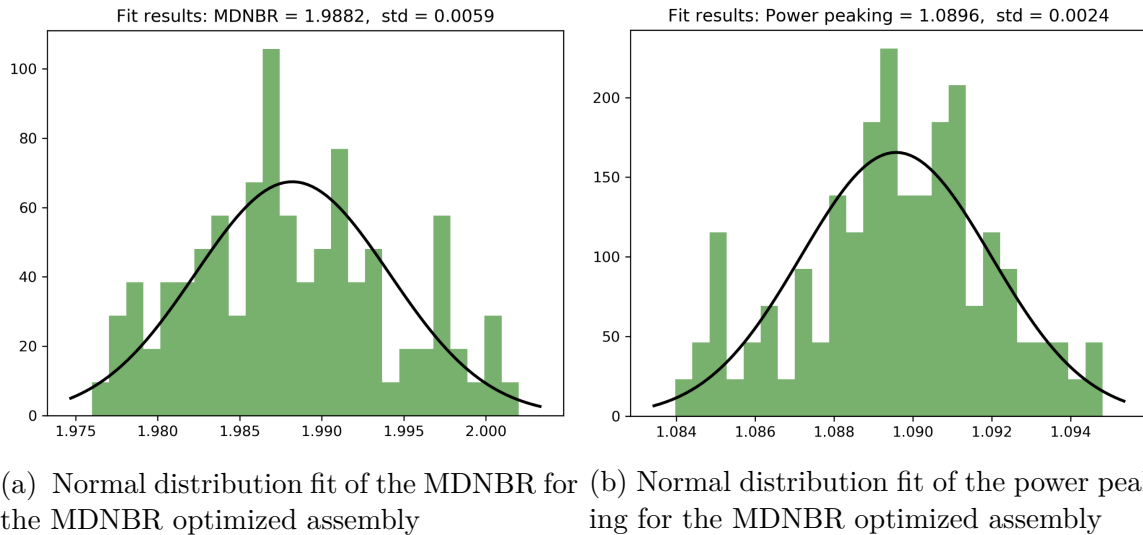


Figure 24: The MDNBR-optimized assembly, Serpent with 10000 particles and 1000 cycles

Then we confirmed that the average difference in pin powers between both sets of pin positions was 0.14% according to one large run of Serpent with a population of 100000 particles per cycle and 1000 cycles. This validates our hypothesis and confirms that using only one Serpent run of pin powers for the whole COBRA optimization is meaningful.

The stability of the pin powers is expected since the new configuration has not changed much from the old one. Indeed, the average displacement (computed only on moving pins) is 0.0039346 cm and the maximum movement is 0.060217 cm. Compared to the pin radius of 0.4287 cm, this means that we moved each pin by 14% of the radius at most. That is quite low and explains why we can't see any visual difference on the heat map which shows pin positions and powers.

We did achieve an improvement of the MDNBR that is clearly present even when ruling out Serpent uncertainties. However, we may still wonder if this improvement of the MDNBR will be significant enough to be visible in another thermohydraulics model or meaningful in real life. Furthermore, since the biggest movement of our pins was of about half a millimetre, we have to question the precision of our mechanical methods for producing an assembly and placing the pins precisely in the desired positions, much less the stability of said positions throughout burnup.

7

A NEW OPTIMIZATION WITH WIMS

Since Monte Carlo methods like Serpent did not yield low enough uncertainties to allow for an optimization in a reasonable amount of time and computing power, we decided to try to model the assembly using a deterministic code. We chose WIMS [17] on Ben Lindley's recommendation because it is able to handle irregular geometries and because the Cambridge nuclear group has some knowledge of the code to be able to help.

7.1 WIMS SETUP

WIMS runs by first computing cross sections for all materials, then modelling transport of neutrons. Modules are called one after the other, each handling an aspect of the simulation. The specific version used here was WIMS 10a.

WIMS first uses the HEAD module to define the materials and their cross sections. We called it on a simple geometry of one fuel pin within a square of size $p_2 = 17/19 * p_0$, the new pitch between the smaller fuel pins (the original pitch is $p_0 = 1.26$ cm). PERSEUS is then called on the same geometry, then PIP does 10 iterations to converge the cross sections.

We then call COND to condense the cross sections into fewer energy groups. We chose partition 22 45 92 135 152 172, which corresponds to the 6-group coarse energy partition compressed from the standard 172 fine-group structure. We also use this module to duplicate some materials : we use the same cross sections as the materials simulated in HEAD to create new materials for the guide tubes (while the original materials are used in the fuel pins). The gap material is composed of very low density Helium gas. PIP has a hard time modelling a gap and won't converge when a gap is present so we simply omit a gap in the HEAD/PERSEUS/PIP geometry and fake its cross section in COND. The cross section of this material does not matter much anyway since its density is so low that no collisions should happen in the gap.

CACTUS is the main module used to simulate the irregular geometry. This is done by modelling each pin individually. First we have to create the assembly geometry itself. We do so by putting in three nodes at (0,0), (0,xmax) and (xmax,xmax), with xmax the half assembly size (10.75 cm). Then we set up edges between all of these nodes and one single element within these edges. For the pins, first we set up the "split" pins which are pins cut in half by a boundary. Then we list the "inserts", pins which are whole pins, entirely inside the geometry boundary. For each pin we give the coordinates of the centre, then the radius of each of the materials : fuel, gap, then cladding as the outer radius for the fuel pins. Following the list of all the pins, we have a list of material numbers with one line per pin and on each line the list of materials in the pin in order of the radii. Then, the keyword "picture" allows us to plot the geometry, as we can see in figure 25. We add in the number of azimuthal angles and their spacing and set up reflective boundary conditions to ensure the correct symmetry is applied.

We have used the CACTUS 2D route without the option RODSUB, since it did not seem to be able to handle irregular geometries. As confirmed by the Amentum help desk, we cannot do any subdivision of rods in the route that we have chosen. This means that there is an inherent uncertainty in our CACTUS 2D tracing. To prevent this, a possible solution would be to plot the geometry in CACTUS 3D instead, which allows CSV type geometry input and sub meshing of any geometry.

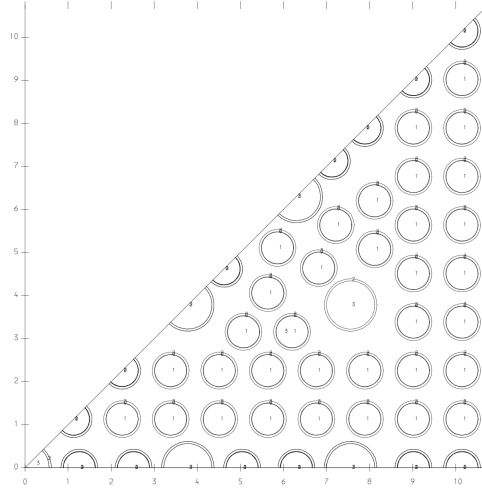


Figure 25: The geometry plotted by the CACTUS 2D module in WIMS

We decided to run WIMS with the following parameters : 32 azimuthal angles, 0.01 cm azimuthal separation and 8 polar angles.

While it is good practice to have the HEAD geometry that generates cross sections as close as possible to the real geometry, HEAD does not accept any irregular geometries, so we are stuck with a one-pin simulation. If we wanted to improve this part in order to get better cross sections we could set up groups of pins which share a similar moderation environment (more moderated close to the assembly edge or to a guide tube, less moderated in the close packed triangles) and model them in separate HEAD calls, with more or less water in the lattice around the pins. This however would take some time to set up, since the pins indexes need to be correctly mapped across the different groups they belong to.

Now that we have set up our simulation, we want to read out our desired results : the power of each pin. First we call the module POWER to normalize the power at 1/8th the desired assembly power. We also add the keyword "fission" to deposit all of the energy at the point of fission. Then the module EDIT is called to use the keyword POWERMAP to get the power of each pin. The creating of a POWERMAP pin list and reading of it after the output is written to the file is quite complicated. We need to give POWERMAP the mesh number of each of the materials that we want to tally on and then invert this process to get back the pin number from the mesh indexes when we want to read the output. This is the last of the modules to be

called in the input file.

We then get two outputs : one is the output file, which includes data about the simulation itself as well as important values to check such as the k_{∞} . The other is the powermap.txt files, which is a list of fission powers for each of the meshes. We then use the function calc P to get the values of pin powers out of this large list, mostly by reversing the process used to set up the POWERMAP section of the input file. We confirm the correct value of total assembly power and add the pin powers to a list in the right format to pass on to COBRA and to the heatmap function which computes power peaking and plots the pin powers.

7.2 WIMS RESULTS AND VALIDATION

We ran this WIMS setup first for the original BL324 pin positions, to compare with our Serpent results. These Serpent results have been validated against Ben Lindley's results and, in general, Monte Carlo methods are considered the reference benchmark. As we can see in table 4 we saw significant differences between WIMS and Serpent. While this can be explained by the approximations made in the WIMS model, as explained above, this is still a disappointing result.

	Serpent	WIMS
k_{∞}	1.43	1.46
Power peaking	1.088	1.025
MDNBR	1.961	1.864

Table 4: Different values as calculated by Serpent and WIMS.

However, the WIMS result can still be of some use. Indeed, as we can see in figure 26, the hottest and coldest pins seem to match between the WIMS and Monte Carlo simulations. The Serpent results are shown again here in figure 27 for comparison (this is the same figure as figure 11). We do not see here the general trend of peaking of the power at the centre of the assembly that we see with the Serpent simulation. This may be because we have computed only one set of cross sections in HEAD, which does not reflect the shift of moderation as we move out from the centre of the assembly.

Seeing that we had an offset in the power peaking between Serpent and WIMS simulations, we were still wondering if the WIMS simulation could capture the trend of the power peaking variations as we move pins.

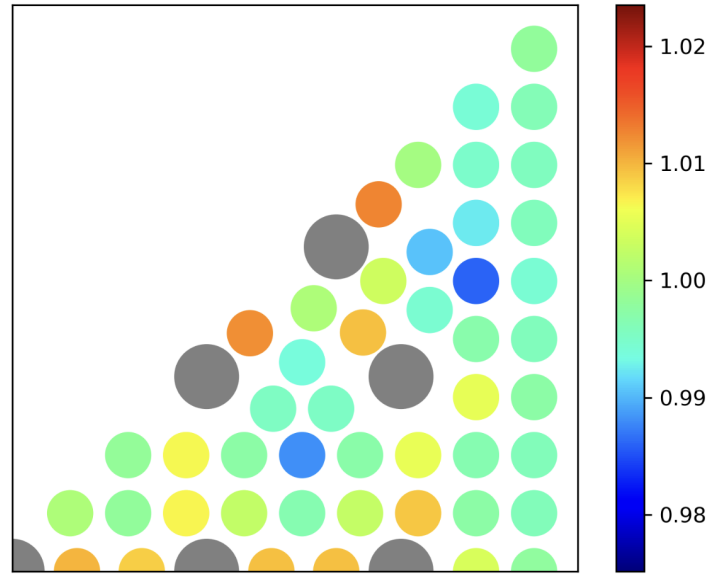


Figure 26: The heatmap for BL324 plotted using WIMS output

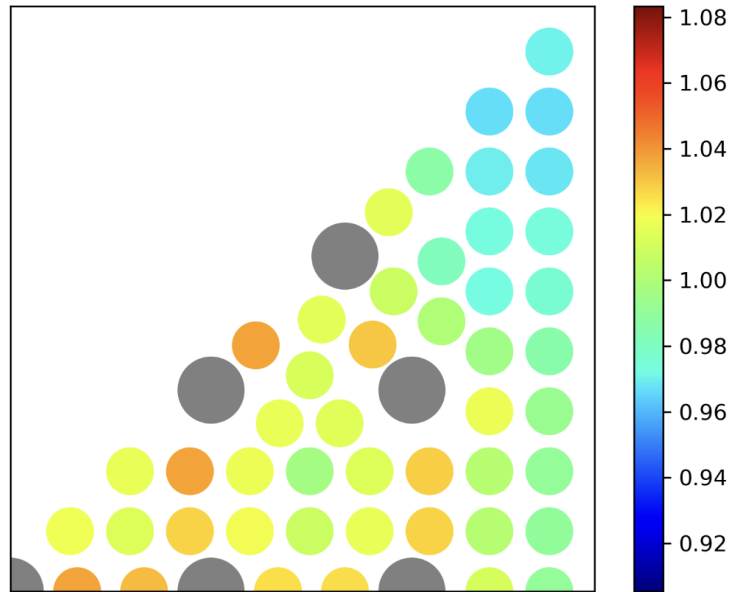


Figure 27: The heatmap for BL324 plotted using Serpent output.

Moving pin 45 In the same manner as our Serpent simulation, we start by moving pin 45 (see figure 18) along the diagonal. We can see in figure 28 the values of power peaking taken from a high fidelity Serpent run (100000 particles per cycle and 1000 cycles) and the values given by WIMS. We have shifted the WIMS curve so that its average matches the Serpent average. This comparison is still interesting since it allows us to consider whether the WIMS

simulation gives the right trend of power peaking even if the actual values are not correct.

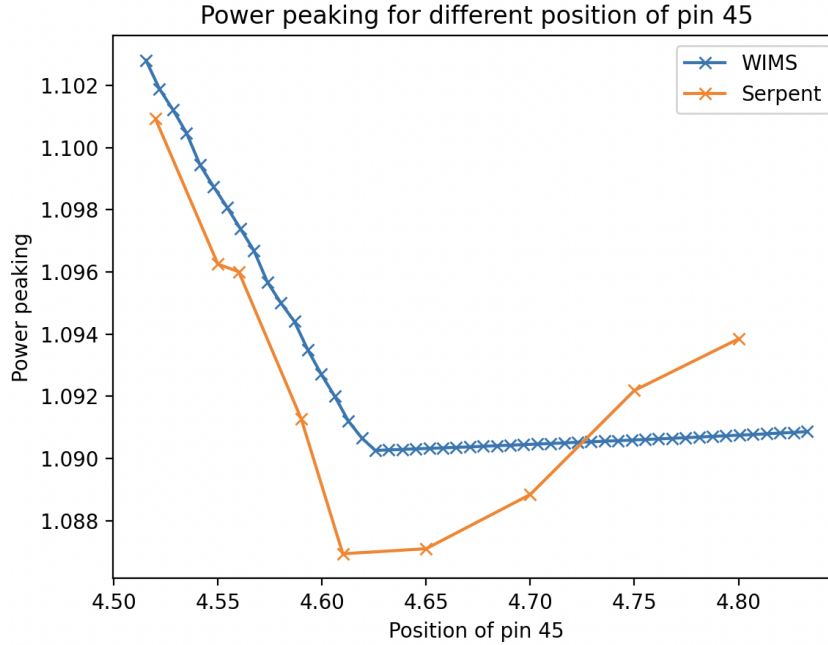


Figure 28: The value of power peaking for each position of pin 45 according to Serpent and WIMS offset to match the average Serpent value

We can see that both curves have a minimum in the same place. This is the original pin 45 positions, so Ben Lindley has identified the best spot for this pin already.

7.3 OPTIMIZATION AND RESULTS

Thanks to the deterministic behaviour of WIMS we used `scipy.optimize.minimize`, the standard Python optimizer. This method only allows deterministic objective functions which can be seen as continuous. The rest of the setup was exactly the same as presented for Serpent, except that a WIMS input file `input` was generated with `BL324 1/8 WIMS maker.py`. WIMS was similarly called with a `subprocess.run` command from the objective function, with special care taken to delete the previous `powermap.txt` file ahead of running WIMS. The process of getting pin powers out of the WIMS output file has been described above and is definitely more complex than for Serpent but was transparently integrated in the optimization loop by modifying the `calc P` function. In general, all care was taken to make the WIMS/Serpent switch a "drop in" solution. The COBRA code in particular does not change at all depending on which neutronics code is giving the pin powers as an input.

7.3.1 • MOVING PIN 32

First, we tried to optimize by only moving one pin. Since the minimum of power peaking when moving pin 45 corresponds to its original position, we have tried to move other pins and see if they are of more interest. Pin 32, the position of which can be seen in figure 29, is interesting since its original position was defined from a regular lattice setup. However, it could move up closer to the guide tube above it.

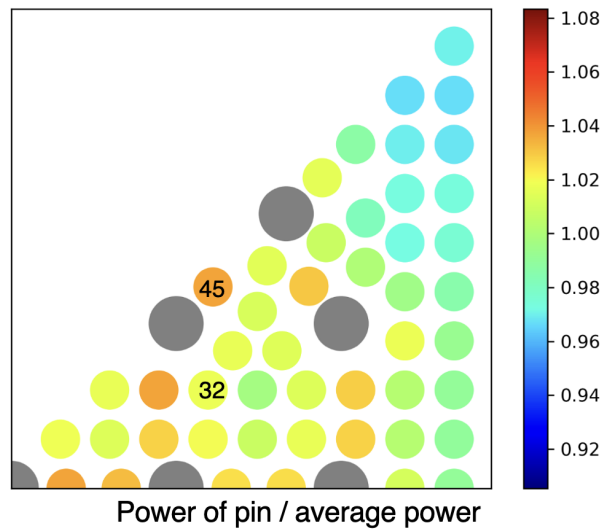


Figure 29: The positions of pins 32 and 45 in the BL324 assembly. The colormap represents power peaking, from the original Serpent run

During this new optimization, we succeeded in finding an optimal position different from the original one. We can see in figure 30 that the optimizer managed to converge to a power peaking value lower than the original one.

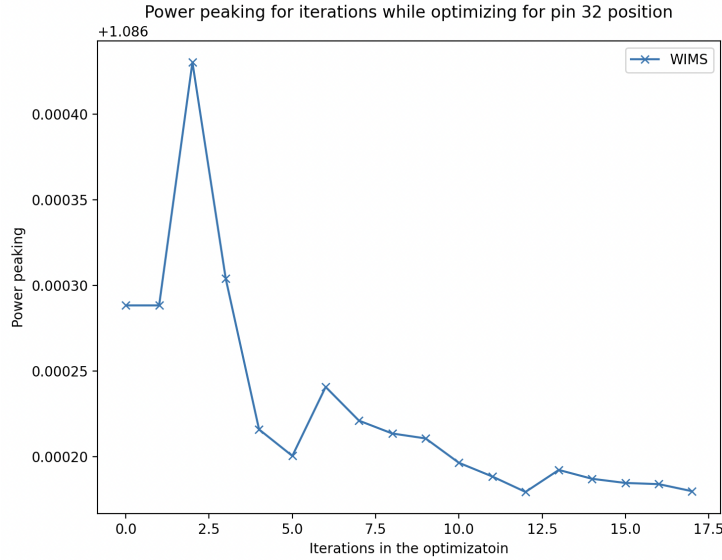


Figure 30: The value of power peaking for each iteration of the optimization. Note +1.086 in the top left corner to read the values of power peaking.

We verify the result by studying the space where pin 32 can move through both WIMS and Serpent. In figures 31a and 31b we can see the power peaking (represented as a colour map) as a factor of the x and y position of the pin 32 centre. The points plotted in the WIMS figure are the points where the optimizer has looked for a minimum and thus run a WIMS simulation. Some of these points were chosen for a Serpent simulation if they seemed to be important for the overall aspect of the trend.

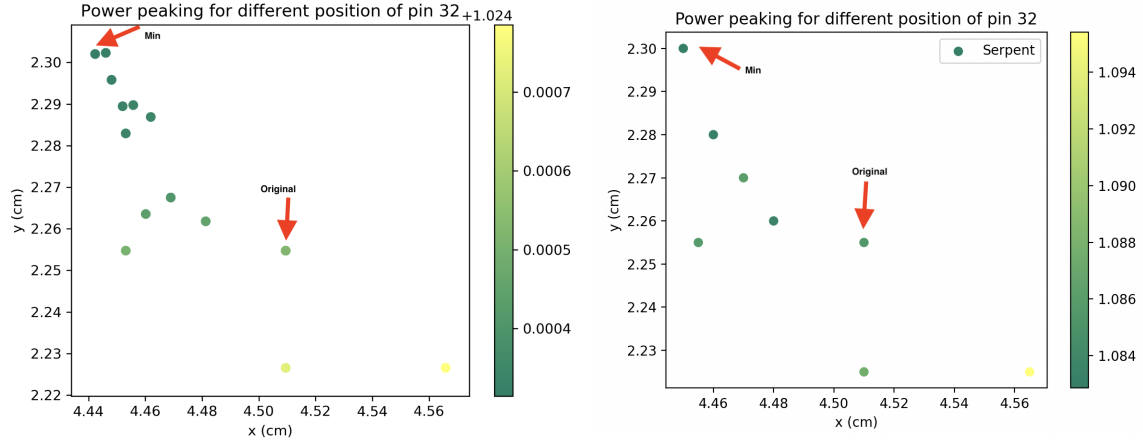
Similarly to the case of pin 45, we can see that the general trend of power peaking depending on pin 32 position is the same for WIMS and Serpent. In particular, the optimal position at the top left of the space studied by the optimizer is indeed a minimum of power peaking according to both Serpent and WIMS.

According to the Serpent run we have the original power peaking at 1.0858 compared to 1.0829 at the new position. This seems to be a clear improvement of the power peaking!

7.3.2 • FULL OPTIMIZATION

Now that we have tried moving one pin around, we wanted to study the movement of all the pins in our 1/8th of an assembly. Similarly to the MDNBR optimization presented above, we allowed the optimizer to move every pin except the ones on the edge of the assembly (right hand column) and the bottom row. The pins on the diagonal were only allowed to move along it while all the other pins had full freedom of movement. The same optimizing method and limits were given. We did manage to find a new set of pin positions for which the power peaking was lower than the original BL324 assembly's according to the WIMS simulation, see figure 32.

However, when we used Serpent to validate this result we do not see as much improvement as expected. As we can see in figure 33, when running Serpent 100 times at 10000 particles per



(a) Power peaking (colormap, note the +1.024 at the top) according to WIMS calculation. (b) Power peaking (colormap) according to Serpent calculation with 10000 particles and 1000 cycles

Figure 31: Power peaking variation with pin 32 position

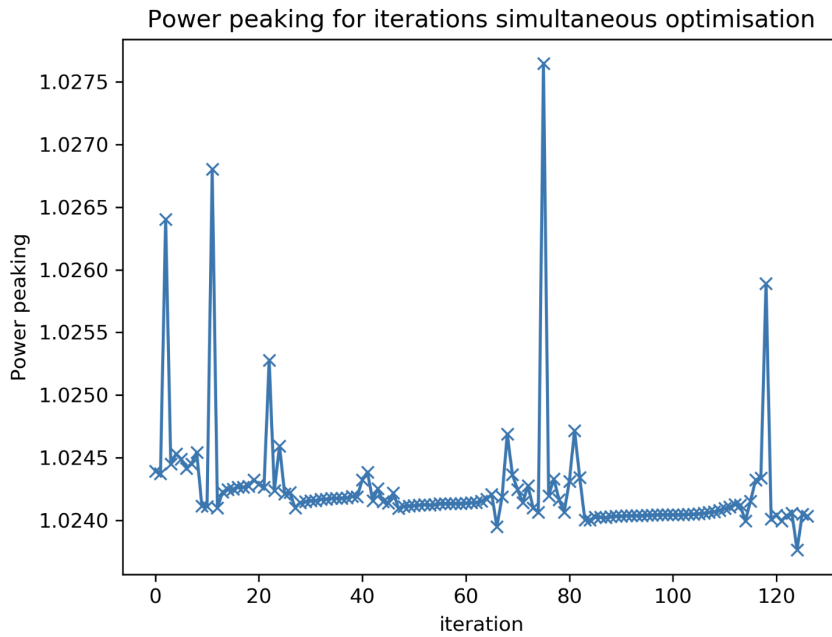


Figure 32: The power peaking for each set of pin positions examined by the optimizer

cycle and 1000 cycles, we see no significant difference in the average power peaking compared to the original BL324 design studied in figure 15a. Indeed, the WIMS-optimized design has a power peaking of 1.0895 with a standard deviation of 0.0017 while the original BL324 has a power peaking of 1.0876 with an almost identical standard deviation of 0.0018. It seems that our new "optimized" design could even have a higher power peaking than the original...

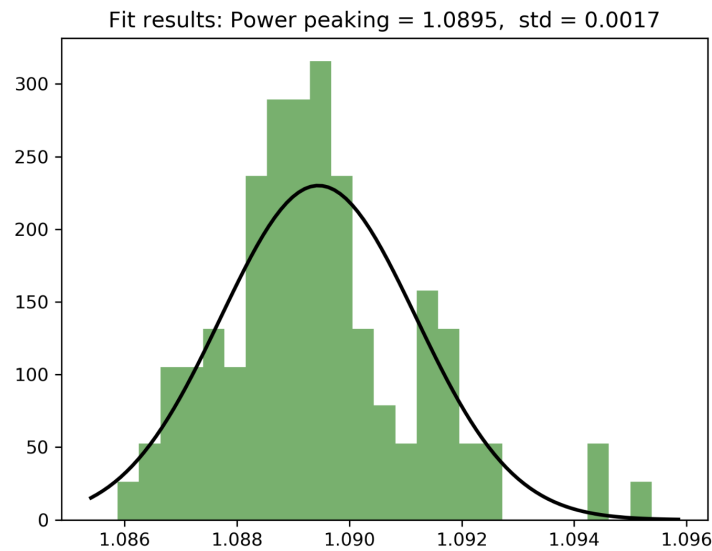


Figure 33: Power peaking values with a normal distribution fit for 100 Serpent runs at 10000 particles per cycle and 1000 cycles. This is using the WIMS-optimized geometry

This means that while on certain pin movements WIMS manages to capture the power peaking trends correctly, it fails to do so when moving all of the pins at once. Thus it seems like the current WIMS setup is not a good way to optimize our irregular assembly.

8

CONCLUSION AND FURTHER STUDY

In this project we have set up a Serpent and WIMS model to compute pin powers in an irregular PWR assembly. We have used a COBRA simulation to compute MDNBR from the pin power values. Then we have used different optimization methods to try and generate a better design. This was done by moving pins while respecting the minimum distance between them.

We first tried to minimize power peaking, which proved difficult with Serpent because of the uncertainties of the Monte Carlo simulation. The deterministic WIMS code did not provide accurate results for power peaking, but the trends were in many cases valid. This allowed us to run an optimization on the position of pin 32 which yielded a better position different from the original BL324 design. However when running an optimization using WIMS where almost all pins could move, we found that the trend was no longer correct. The new assembly found by the optimizer was not significantly better than the original BL324 design.

Secondly we optimized the irregular assembly to get the highest MDNBR possible. This was easier to run since we used the assumption that pin powers will not vary as we move pins around by small amounts. We found an optimal position with this assumption and when running a full simulation including the real pin powers at the new pin positions we did see a clear improvement of MDNBR.

While work on this project has been difficult because of the computational expense and the difficulty of modelling irregular assemblies, some results have been obtained. The MDNBR optimization is especially meaningful but some progress has also been made towards a power peaking optimization. It seems that the design Ben Lindley came up with is very close to optimal and only small changes of pin positions are allowed around this first set of positions.

Further work on this project would be interesting. For this purpose, we have left the codes used in this project available for future students in a Sharepoint folder and have tried to comment them properly.

First, there is room for more work to conclude on the possibility of optimizing the BL324 assembly by moving pins. As we have explained while detailing the WIMS setup, a first avenue of research would be to improve the WIMS simulation. This could be done through better meshing for transport, which would be possible if we used the CACTUS 3D module, or through better cross sections, for example by separating more moderated pins from less moderated ones when calculating cross sections. It would also be possible to simply give more computing power to the Serpent model we have set up here. It remains to be seen whether that would yield results, since there is still no certainty that a much better design than the one Ben Lindley proposed can be found.

Since we have seen that improving both power peaking and MDNBR are interesting objectives, one could think of a way to improve MDNBR and power peaking together, maybe using a composite objective function.

We have also started working on studying a set of different optimization algorithms. One

issue we faced during this project was to find an algorithm capable of suggesting non overlapping solutions without staying too close to Lindley's design. It was suggested that we try a genetic algorithm, on which we did some initial work. However the choice of an initial population is difficult. A random initial population means that the algorithm fails to find any non-overlapping design, while starting from Lindley's design tends to have the same bias towards finding a local minimum similar to Lindley's solution. Improving a genetic algorithm for use in this situation would be an interesting study to run. However it might not be the best suited to this task and other algorithms could also be studied.

Secondly, if a better assembly is found, one would need to test for all of the constraints that are not currently evaluated by the optimization loop. This includes mostly the evolution of reactivity with burnup and control rod worth. These elements were not included in the current optimization process because they should not pose any problem, since there is no reason the behaviour of our assembly should be different from BL324 or even a traditional assembly.

Then, there are possibilities for further scope of work. One could study other ways of optimizing the assembly rather than only moving pins around. Such further optimization could include having different enrichments for different pins or burnable poisons in some pins. This would allow us to further reduce power peaking.

In these cases, it might not always be possible to remove the study of the evolution of reactivity with burnup and control rod worth from the optimization loop. Indeed, in some cases such as the presence of burnable absorbers, the maximum power peaking is not necessarily at the beginning of the fuel cycle. In this case a burnup study must be included in the optimization itself, in order to use the maximum power peaking throughout the cycle as the figure of merit in the optimization.

Since Ben Lindley and his group have been working on modelling full cores of the BL324 assemblies, and would like to study transition cores with both BL324 and traditional assemblies, it could be useful to see where a possible optimized design would fit into this. Possibly these models could be adapted to a new, optimized 324 pin design.

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